Tutorial-6

Property Features and prices

Bedrooms (x_1)	Age (x_2)	Price (100000s) (y)
3	5	2
3	3	3

1. write the ridge regression cost function
2. calculate weight updates.

Cost Function:

$$J = \|y - xw\|^2 + \lambda \|w\|^2$$

Slopes

$m_1 \rightarrow x_1$

$m_2 \rightarrow x_2$

Linear equation

$2 = c + m_1 \cdot 3 + m_2 \cdot 5$

$3 = c + m_1 \cdot 3 + m_2 \cdot 3$

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix}}_{x} \underbrace{\begin{pmatrix} c \\ m_1 \\ m_2 \end{pmatrix}}_{w}$$

$$w = (x^T x + \lambda I)^{-1} (x^T y)$$

$$x^T = \begin{bmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{bmatrix}$$

$\boxed{\because \lambda = 1}$

$$x^T x + \lambda I = \begin{bmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{bmatrix} + \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+1 & 3+3 & 5+3 \\ 3+3 & 9+9 & 15+9 \\ 5+3 & 15+9 & 25+9 \end{bmatrix} + 1 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$X^TX+\lambda I = \begin{bmatrix} 3 & 6 & 8 \\ 6 & 19 & 24 \\ 8 & 24 & 35 \end{bmatrix}$$

$$(X^TX+\lambda I)^{-1} = \begin{bmatrix} \frac{89}{95} & \frac{-18}{95} & \frac{-8}{95} \\ \frac{-18}{95} & \frac{41}{95} & \frac{-24}{95} \\ \frac{-8}{95} & \frac{-24}{95} & \frac{21}{95} \end{bmatrix}$$

$$X^Ty = \begin{bmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 15 \\ 19 \end{bmatrix}$$

$$(X^TX+\lambda I)^{-1}(X^Ty) = \begin{bmatrix} \frac{89}{95} & \frac{-18}{95} & \frac{-8}{95} \\ \frac{-18}{95} & \frac{41}{95} & \frac{-24}{95} \\ \frac{-8}{95} & \frac{-24}{95} & \frac{21}{95} \end{bmatrix} \begin{bmatrix} 5 \\ 15 \\ 19 \end{bmatrix}$$

$$= \begin{bmatrix} 0.2421 \\ 0.7263 \\ -0.0105 \end{bmatrix} = \begin{bmatrix} c \\ m_1 \\ m_2 \end{bmatrix}$$

$$\hat{y} = 0.2421 + 0.72x_1 - 0.01x_2$$

$$\hat{y}_1 = 0.2421 + 0.72(3) - 0.01(5) = 2.35$$

$$\hat{y}_2 = 0.2421 + 0.72(3) - 0.01(3) = 2.37$$

3. Cost Function = $J = \|y - xw\|^2 + \lambda \|w\|^2$

$$X = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 13 & 3 \end{bmatrix} \quad w = \begin{bmatrix} 0.242 \\ 0.72 \\ 0.010 \end{bmatrix}$$

$$Xw = \begin{bmatrix} 2.352 \\ 2.372 \end{bmatrix}$$

$$Y - Xw = \begin{bmatrix} -0.35 \\ 0.63 \end{bmatrix}$$

$$\|Y - Xw\|^2 = (-0.35)^2 + (0.57)^2 = 0.52$$

$$\|w\|^2 = c^2 + m_1^2 + m_2^2$$

$$= (0.2421)^2 + (0.72)^2 + (-0.010)^2$$

$$= 0.577$$

$$\|Y - Xw\|^2 + \lambda \|w\|^2 = 0.52 + 0.57$$

Cost function = $\underline{\underline{1.09}}$